

Cancer	Rank	Peak (cm ⁻¹)	Representative assignment	Candidate assignments	Feature importance
LUN	1	1597	Purine C=C/C=N	Purine C=C/C=N; adenine/guanine; aromatic amino-acid mode	10.62
LUN	2	795	Hippuric acid	Hippuric acid; protein C-C/C-N mode; ring breathing mode	8.46
LUN	3	618	C-S stretching	C-S stretching; protein sulfur-containing residues; creatinine-related mode	8.23
LUN	4	1000	Phenylalanine	Phenylalanine; urea; aromatic ring breathing	6.89
CRC	1	795	Hippuric acid	Hippuric acid; protein C-C/C-N mode; ring breathing mode	9.50
CRC	2	1597	Purine C=C/C=N	Purine C=C/C=N; adenine/guanine; aromatic amino-acid mode	7.63
CRC	3	618	C-S stretching	C-S stretching; protein sulfur-containing residues; creatinine-related mode	6.17
CRC	4	1000	Phenylalanine	Phenylalanine; urea; aromatic ring breathing	5.69
BLC	1	1597	Purine C=C/C=N	Purine C=C/C=N; adenine/guanine; aromatic amino-acid mode	10.78
BLC	2	618	C-S stretching	C-S stretching; protein sulfur-containing residues; creatinine-related mode	10.71
BLC	3	895	Uric acid	Uric acid; nucleic-acid backbone/ring mode; C-C skeletal mode	10.27
BLC	4	1000	Phenylalanine	Phenylalanine; urea; aromatic ring breathing	10.12
PRO	1	1597	Purine C=C/C=N	Purine C=C/C=N; adenine/guanine; aromatic amino-acid mode	13.99
PRO	2	1651	Amide I / protein	Amide I / protein; C=C lipid stretching; water/protein carbonyl mode	13.39
PRO	3	1449	CH2 deformation / lipid	CH2 deformation / lipid; protein CH2/CH3 bending; collagen-related mode	9.32
PRO	4	1231	Amide III / protein	Amide III / protein; nucleic-acid base mode; CH bending	9.29
OVA	1	1597	Purine C=C/C=N	Purine C=C/C=N; adenine/guanine; aromatic amino-acid mode	12.02
OVA	2	1651	Amide I / protein	Amide I / protein; C=C lipid stretching; water/protein carbonyl mode	9.47
OVA	3	618	C-S stretching	C-S stretching; protein sulfur-containing residues; creatinine-related mode	9.03
OVA	4	895	Uric acid	Uric acid; nucleic-acid backbone/ring mode; C-C skeletal mode	8.85
PAN	1	795	Hippuric acid	Hippuric acid; protein C-C/C-N mode; ring breathing mode	8.73
PAN	2	1597	Purine C=C/C=N	Purine C=C/C=N; adenine/guanine; aromatic amino-acid mode	8.04
PAN	3	618	C-S stretching	C-S stretching; protein sulfur-containing residues; creatinine-related mode	6.31
PAN	4	1651	Amide I / protein	Amide I / protein; C=C lipid stretching; water/protein carbonyl mode	6.17